

Project Proposal

Project Title: 3D Modeling and Interactive Rendering of AnimeSword)

Group No: 15

1. Topic & Objective Our team proposes to design, model, and render a 3D representation of the signature weapon used by Gen Narumi from the series *Kaiju No. 8*. This weapon is a complex bayonet-style rifle that combines industrial mechanical design with glowing edged, energy-based elements.

2. Description of the Idea & Technical Approach We will construct the model procedurally using geometric primitives and manual vertex specification. The project will be divided into three main technical components:

- **Complex Geometric Modeling:** Unlike simple shapes, this weapon features a jagged, non-uniform blade and a cylindrical gun barrel. We will implement custom algorithms to define 2D profiles for the blade and "extrude" them into 3D volumes by mathematically generating the connecting side walls (quads) and front/back faces.
- **Hierarchical Structure:** The weapon is an assembly of distinct parts (Barrel, Main Blade, Gun body, Secondary Blade, Handle).
- **Lighting and Visual Effects:** To make the weapon look realistic, we will use different settings for its materials. This includes adjusting the surface details to make the metal body look shiny and reflective, and using light settings to make the magenta energy on the edges glow.

3. Expected Final Output The final deliverable will be an interactive 3D application with the following characteristics:

- **Visual Fidelity:** The rendered object will appear as a cohesive, solid 3D model with a dark, matte metallic finish contrasted by bright, glowing energy strips along the blade's cutting edge.
- **Interactivity:** The application will feature a user-controlled camera system allowing for:
 - **Orbiting:** Rotating the model on X and Y axes to inspect details from all angles.
 - **Panning & Zooming:** Inspecting specific parts of the geometry.
 - **Animation:** An animation of the weapon extending and retracting